LEVEL UP ACADEMY

Class 10 | GSEB | Science

ANSWER KEY & MARKING SCHEME

Ch 5 Test (GSEB pattern): Life Processes

Chapter / Topic: Life Processes

For teacher use only. Do not distribute to students.

SECTION A

Q1. The kidneys in human beings are a part of the system for: **[1]**

Answer: (a) excretion

Q2. Which of the following shows holozoic nutrition? **[1]**

Answer: (c) Amoeba

Q3. The xylem in plants is responsible for the transport of: **[1]**

Answer: (b) water (and dissolved minerals)

Q4. The finger-like projections on the inner lining of the small intestine are called: **[1]**

Answer: (c) villi

Q5. The autotrophic mode of nutrition requires: **[1]**

Answer: (d) all of the above

Q6. Which blood vessel carries deoxygenated blood from the heart to the lungs? **[1]**

Answer: (a) Pulmonary artery

Q7. In aerobic respiration, the breakdown of pyruvate to give carbon dioxide, water and energy takes place in the: **[1]**

Answer: (d) mitochondria

Q8. The opening and closing of stomata is caused by: **[1]**

Answer: (b) a change in the water content of the guard cells

Q9. Which blood vessel carries oxygenated blood from the lungs to the heart? **[1]**

Answer: (a) Pulmonary vein

SECTION B

Q10. How are the alveoli designed to maximise the exchange of gases? **[2]**

Answer: The alveoli are balloon-like structures that give a very large surface area. Their walls are very thin, and they have an extensive network of blood capillaries, so gases diffuse easily.

- Large surface area: 1
- Thin walls + capillary network: 1

Q11. Why do aquatic organisms breathe much faster than terrestrial organisms? **[2]**

Answer: The amount of oxygen dissolved in water is much lower than the amount of oxygen in air. So aquatic organisms must breathe faster to get enough oxygen.

- Less dissolved oxygen in water: 1
- So faster breathing: 1

Q12. Why is it necessary to separate oxygenated and deoxygenated blood in mammals and birds? **[2]**

Answer: Mammals and birds need a lot of energy, especially to maintain a constant body temperature. Keeping the two kinds of blood separate gives a highly efficient supply of oxygen to the body.

- High energy need / constant body temperature: 1
- Efficient oxygen supply: 1

SECTION C

Q13. Answer the following about photosynthesis: **[3]**

(i) $6CO_2 + 12H_2O \rightarrow C_6H_{12}O_6 + 6O_2 + 6H_2O$ (in the presence of sunlight and chlorophyll) [1]

(ii) (1) Chlorophyll absorbs light energy. (2) Light energy is converted to chemical energy, and water molecules are split into hydrogen and oxygen. (3) Carbon dioxide is reduced to carbohydrates. [2]

- Three events: 2 (any two correct: 1)

Q14. Answer the following about the human heart: **[3]**

(i) Four: two atria and two ventricles. [1]

(ii) The left ventricle (through the aorta). [1]

(iii) They make sure blood does not flow backwards when the atria or ventricles contract. [1]

SECTION D

Q15. When the kidneys fail, an artificial kidney (dialysis machine) can be used. It contains a number of tubes with a semi-permeable lining, suspended in… **[4]**

(i) Urea (and uric acid). [1]

(ii) So that the wastes can diffuse from the blood, where their concentration is higher, into the fluid. [1]

(iii) In dialysis there is no reabsorption. A healthy kidney selectively reabsorbs useful substances like glucose, amino acids, salts and water back into the blood. [2]

- No reabsorption in dialysis: 1
- Kidney reabsorbs useful substances: 1